

Sparse Identification-Enabled Data-Driven Modeling for Nonlinear Dynamics of Microgrids

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Abstract—Considering the widespread integration of power-electronic-interfaced distributed energy resources (DERs) in microgrids, the system's inertia is highly reduced. Thus, microgrids are sensitive to disturbances. To get the situational awareness of transient dynamics of microgrids when they face severe disturbances, a data-driven modeling algorithm, Sparse Identification of Non-Linear Dynamics (SINDy), is discussed in this paper. The governing equation of microgrids can be identified through SINDy, which determines the nonlinear dynamics of the system. Numerical examples test and verify the data-driven model based on SINDy can successfully capture the transient dynamics of microgrids. Simulation results have validated the data-driven model is reliable and very close to the actual test system. Therefore, it provides a potent tool to analyze the system's transient dynamics when an accurate system's model is unavailable.

Index Terms—Sparse identification, microgrids, distributed energy resources (DERs), data-driven, nonlinear dynamics, system identification.

I. INTRODUCTION

Microgrid is defined as a group of interconnected loads and distributed energy resources (DERs) that acts as a single controllable entity with respect to the power grid [1]. It is recognized as a powerful paradigm because of its capability to significantly improve the flexibility and resilience of low- or medium-voltage distribution networks and facilitate the high penetration of DERs [2] such as wind generation, photovoltaic and Energy Storage Systems (ESS) [3]. Since the transient characteristics of microgrids are very important for the planning, operation, and control of the system, it is critical to accurately model the transient dynamics. Conventionally, the dynamic mathematical model for a microgrid is presented by a set of Differential-Algebraic Equations (DAEs), where the differential equations are used to model the system dynamics and the algebraic equations are introduced as constraints for the state variables [4]. The high fidelity mathematical models are especially useful while applying new control strategies and performing stability analysis on the system [5], [6].

Multiple studies have been undertaken to define a detailed dynamical model for microgrids. However, it is still a challenging problem due to the following three major reasons. First, the use of available detailed dynamic models for each DER unit requires large computational power and becomes very difficult for the study of large-scale microgrids [7], [8]. Second, the process of modeling a microgrid that is based on dynamical equations is complex and convoluted as

there are multiple DER units, loads and power electronic components, each with their mathematical model contributing to the overall dynamic model of the microgrid system. Third, the addition of new components/ units to an existing system would completely change the system model and modifications have to be introduced to the existing dynamics to accurately represent the new system. Modifying the dynamical equations to follow the changes every single time would be a challenging process. Considering the widespread installation of the smart metering system, the system's operational data is available. One constructive question is: *Could the transient dynamics of microgrids be identified via systems' operational data, namely developing a data-driven model?*

Identifying the dynamics of the system using a data-driven model is a concept where the emphasis is placed on the data that can be easily obtained from the actual physical system rather than a congregate of multiple equations pertaining to different elements in the system. Additionally, the use of data-driven model helps accurately determine the system dynamics to overcome possible unexpected inaccuracies, since the changes in the system state would immediately train the data-driven model and modify the dynamics accordingly. Nonlinear models based on machine learning concepts, e.g., neural network based system identification [9], are increasingly used due to advances in computing power and can be applied to several complex systems [10]. Deep reinforcement learning has also been combined with Model Predictive Control (MPC) [11], yielding impressive results in the large data limit. However, machine learning algorithms often lack interpretability and may suffer from overfitting issue [12]. To overcome the effect of overfitting in system identification, sparse identification of non-linear dynamics (SINDy) algorithm has been proposed in [13]. Exploiting the sparsity of the non-linear equations in the space of all possible functions, the governing equations can be determined and trained to follow the true system [14]. The method is computationally efficient and is a sustainable alternative for online identification in response to rapid system changes [15]. A model-free method using Koopman operator approximation has been proposed in [16] for power system models - Kundur two area system and IEEE 39-bus test system. This method obtains a linear approximation of the non-linear system and does not depend on the state variables as opposed to the SINDy method. The advantage of the SINDy method is the ability of the SINDy method to adapt to new

test cases in real-time. This paper focuses on leveraging the concept of SINDy-based system identification to develop a data-driven model for nonlinear dynamic microgrids.

The contributions of the presented work are stated below:

- A data-driven model, namely Ordinary Differential Equations (ODEs), is developed for nonlinear dynamic microgrids by performing the SINDy algorithm on the measurement data obtained from the system. The ODEs form simplifies the computation of the original DAEs, which is numerically more challenging and more severe than the ODE form.
- Leveraging the fact that the nonlinear system equations representing the microgrid model is sparse in the space of all possible candidate mathematical functions, the governing equations of microgrids can be efficiently identified through the SINDy method.
- The data-driven model can be easily adaptive according to the measurement data, as the model development does not make any assumptions related to the actual physical microgrids and relies only on the measurement data.

The remainder of this paper is organized as follows. Section II introduces the mathematical concepts of the SINDy algorithm. In Section III, tests on a typical nonlinear microgrid system have verified the effectiveness of the SINDy method in modeling the system in a data-driven manner. Conclusions are drawn in Section IV.

II. SPARSE IDENTIFICATION OF NONLINEAR DYNAMICS

The SINDy approach is used to discover the governing equations for dynamical systems by purely using the data measured physically from the system of interest. The different values of system variables obtained over time can be used to discover the nonlinear dynamic system based on sparse regression techniques [13], [14]. The fact that governing equations of most dynamic systems are sparse in a high dimensional function space implies that the physical system is only defined by a few relevant terms of the system. This property makes SINDy a powerful tool for system identification. Generally, a nonlinear dynamic system can be modeled in (1).

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t)), \quad (1)$$

where $\mathbf{x}(t) = [x_1(t), x_2(t), \dots, x_n(t)]^T \in \mathbb{R}^n$ is the set of state variables in the system at the time step t . $\mathbf{f}(\mathbf{x}(t))$ represents the nonlinear function defining the governing equations of the system in terms of the state variables. There are three steps to determine the governing equations $\mathbf{f}(\mathbf{x}(t))$ from the measurement data, as summarized below.

A. Preparing Data for Identifying the Governing Equations

The data of the state vectors $\mathbf{x}(t)$ over time t is collected for calculating their derivatives $\dot{\mathbf{x}}(t)$ analytically or numerically. One way $\dot{\mathbf{x}}(t) \approx \frac{\Delta \mathbf{x}}{\Delta t}$ can be used to approximate its numerical value, where Δt is the sampling time.

Considering the samples of measurement, the data of the n state variables $\mathbf{X}$ sampled over the given time period $t = [t_1, t_2, \dots, t_m]$ and its derivatives $\dot{\mathbf{X}}$ obtained through

the direct measurement from the system or through analytical computation can be represented using the following matrices given in (2) and (3). In microgrids, the system's state variables, bus voltages, and currents will be measured to build the data matrices for system identification.

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}^T(t_1) \\ \mathbf{x}^T(t_2) \\ \vdots \\ \mathbf{x}^T(t_m) \end{bmatrix} = \begin{bmatrix} x_1(t_1) & x_2(t_1) & \cdots & x_n(t_1) \\ x_1(t_2) & x_2(t_2) & \cdots & x_n(t_2) \\ \vdots & \vdots & \ddots & \vdots \\ x_1(t_m) & x_2(t_m) & \cdots & x_n(t_m) \end{bmatrix} \quad (2)$$

$$\dot{\mathbf{X}} = \begin{bmatrix} \dot{\mathbf{x}}^T(t_1) \\ \dot{\mathbf{x}}^T(t_2) \\ \vdots \\ \dot{\mathbf{x}}^T(t_m) \end{bmatrix} = \begin{bmatrix} \dot{x}_1(t_1) & \dot{x}_2(t_1) & \cdots & \dot{x}_n(t_1) \\ \dot{x}_1(t_2) & \dot{x}_2(t_2) & \cdots & \dot{x}_n(t_2) \\ \vdots & \vdots & \ddots & \vdots \\ \dot{x}_1(t_m) & \dot{x}_2(t_m) & \cdots & \dot{x}_n(t_m) \end{bmatrix} \quad (3)$$

B. Establishing Candidate Nonlinear Functions

A library of candidate nonlinear functions $\Theta(\mathbf{X})$ will be built, consisting of vectors of constant values, polynomials and trigonometric terms. This library of possible nonlinear time series function space will help build the pool data of candidate functions, with an example given in (4).

$$\Theta(\mathbf{X}) = \begin{bmatrix} \mathbf{1} & \mathbf{X} & \mathbf{X}^{\mathbf{P}^2} & \cdots & \sin(\mathbf{X}) & \cos(\mathbf{X}) & \cdots \end{bmatrix} \quad (4)$$

The candidate function matrix is chosen such that the number of candidate functions (columns of the matrix $\Theta(\mathbf{X})$) is less than the data samples (rows of the matrix $\Theta(\mathbf{X})$). The sparsity of the dynamic equations of the microgrid system enables us to obtain proper tracking of the original system with lesser number of functions; and hence, this restricted basis (with polynomials and trigonometric functions) can be chosen to represent the $\Theta(\mathbf{X})$. In (4), the polynomial terms are given by $\mathbf{X}$, $\mathbf{X}^{\mathbf{P}^2}$, $\dots$, $\mathbf{X}^{\mathbf{P}^n}$, where the polyorder, i.e., the order of the polynomial functions, determines the range of values for the higher order polynomial functions required for performing the sparse regression analysis. Similarly, the trigonometric terms represented by $\sin(\mathbf{X})$, $\cos(\mathbf{X})$, $\mathbf{X} \sin(\mathbf{X})$, $\mathbf{X} \cos(\mathbf{X})$, $\dots$, $\mathbf{X}^n \sin(\mathbf{X})$, $\mathbf{X}^n \cos(\mathbf{X})$ determine the range of values for all the trigonometric functions required for performing the sparse regression analysis. Samples of the quadratic polynomial function $\mathbf{X}^{\mathbf{P}^2}$ and the first order trigonometric function $\sin(\mathbf{X})$ is explicitly given as below.

$$\mathbf{X}^{\mathbf{P}^2} = \begin{bmatrix} x_1^2(t_1) & x_1(t_1)x_2(t_1) & \cdots & x_2^2(t_1) & \cdots & x_n^2(t_1) \\ x_1^2(t_2) & x_1(t_2)x_2(t_2) & \cdots & x_2^2(t_2) & \cdots & x_n^2(t_2) \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ x_1^2(t_m) & x_1(t_m)x_2(t_m) & \cdots & x_2^2(t_m) & \cdots & x_n^2(t_m) \end{bmatrix} \quad (5)$$

$$\sin(\mathbf{X}) = \begin{bmatrix} \sin(x_1(t_1)) & \sin(x_2(t_1)) & \cdots & \sin(x_n(t_1)) \\ \sin(x_1(t_2)) & \sin(x_2(t_2)) & \cdots & \sin(x_n(t_2)) \\ \vdots & \vdots & \ddots & \vdots \\ \sin(x_1(t_m)) & \sin(x_2(t_m)) & \cdots & \sin(x_n(t_m)) \end{bmatrix} \quad (6)$$

C. Performing the Sparse Regression Analysis

The sparse regression analysis is then carried out on the pool data to identify the function $f(\mathbf{x}(t))$ in (1). For microgrids, only a few of the nonlinear terms will appear in the function $f(\mathbf{x}(t))$, making it sparse in the space of all possible candidate functions. Leveraging the idea that the function is sparse in nature, we apply a regression method to identify the linear combination of these nonlinear terms that represent $\dot{\mathbf{X}}$.

The sparse vector coefficients identified by using regression method can be given by $\Xi = [\xi_1 \ \xi_2 \ \cdots \ \xi_n]$ and then $\dot{\mathbf{X}}$ is identified to be,

$$\dot{\mathbf{X}} = \Theta(\mathbf{X})\Xi. \quad (7)$$

Multiple regression methods can be used to identify the vector Ξ , e.g., Ordinary Least Squares (OLS) regression, Ridge Regression, Least Absolute Shrinkage and Selection Operator (LASSO) method, etc. Thus, a data-driven model can be determined, i.e., the function $f(\mathbf{x}(t))$, only with the time series value of the system's variables, which in turn gives the governing equations of the system.

D. Applications to Microgrids

Mathematically, the dynamic microgrid system can be modeled by a set of differential-algebraic equations (DAEs) when the power-electronic interfaces are modeled by averaged switch modeling, as given in (8).

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}, \mathbf{y}) \\ \mathbf{0} = \mathbf{G}(\mathbf{x}, \mathbf{y}), \end{cases} \quad (8)$$

where $\mathbf{x} \in \mathbb{R}^n$ is the state variable vector, e.g., state variables in the controller of power-electronic interfaces, which are detailed in Section III; $\mathbf{y} \in \mathbb{R}^m$ is the algebraic variable vector, e.g., bus voltage amplitude and angle. Both the differential equation and the algebraic equation given in (8) depends on the state variables $\mathbf{x}$; and thus, (8) can be converted to ODE, as discussed below. The algorithm has been designed in such a way to combine the time series data of states and output variables observed in the actual model to act as the constraint to define the dynamic equations of the identified system. The final identified system would be a pure differential equation that could track both the states and the outputs. The basic system equations have small order polynomials and sinusoidal terms to define the inverter and control equations. Hence, the candidate function has been chosen to as given in (4).

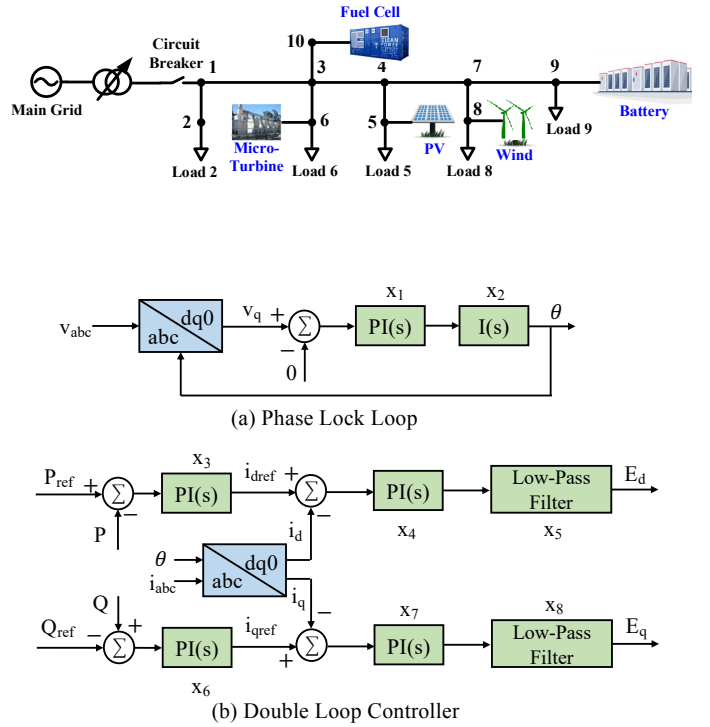


Fig. 2. The control system for DER units.

III. NUMERICAL EXAMPLES

A typical microgrid system shown in Fig. 1 is used to test and validate the feasibility of SINDy approach in establishing a data-driven model for determining the dynamic operations of the nonlinear system [1]. The test system operates in the islanded mode, i.e., Circuit Breaker is open; thus, the system is sensitive to disturbances. Since the Micro-turbine has a large power capability, in the test, bus 6 is set as the swing bus. Double loop controller shown in Fig. 2 is used to control the DERs, with PQ control given as an example. There are a total of 40 state variables. All the five loads are constant power loads. The data-driven model is developed via the SINDy method by using the system's states data and the output voltage data. The feasibility of the identified governing equations is analyzed from the following three aspects.

A. Comparison of Time Domain Simulations

The active power of Load 6 increases by 10% at $t = 0.0$ s. Fig. 3 shows the comparison of the state variables of Micro-turbine in the test microgrid. The states x_1, x_2, x_4, x_7 are plotted for both the true system (DAE form) and the identified system (ODE form). The physical meaning of the different states (i.e. $x_1, x_2, \dots, x_8$) are mentioned in Fig. 2. The comparison in Fig. 3 proves that the SINDy algorithm can be used to effectively track the true system. In order to understand the robustness of the data-driven model to large disturbances, the active power load was varied from 10% to 50% at $t = 0.0$ s and satisfactory tracking performance was observed in the data driven model. Note that the state variable

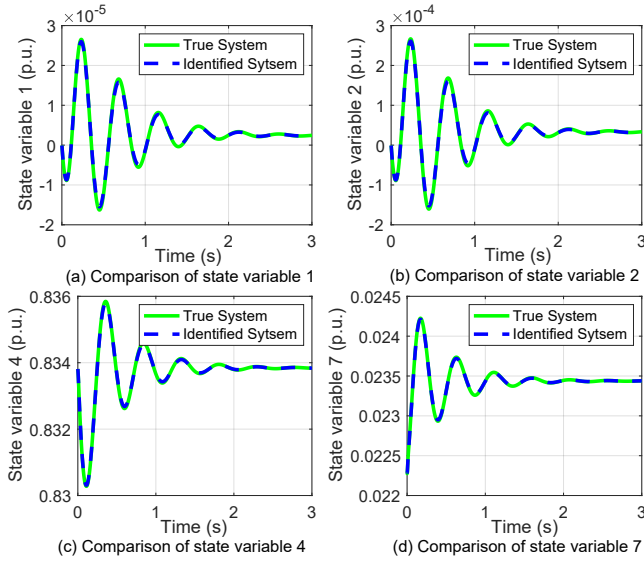


Fig. 3. Comparison of state variables x_1 , x_2 , x_4 , and x_7 between the true system and identified system for Micro-turbine.

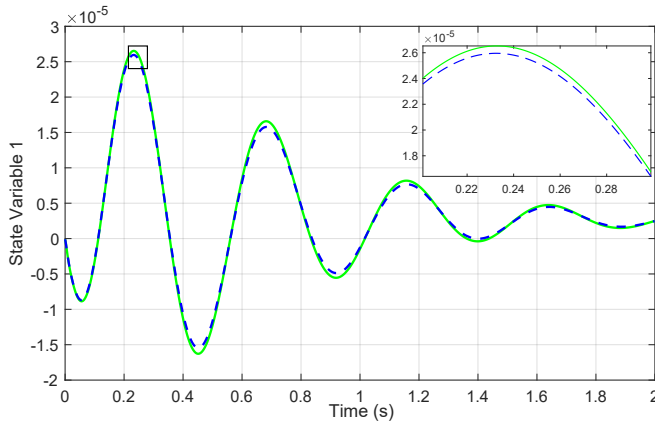


Fig. 4. Zoom in details of State Variable 1 for Micro-turbine.

comparisons are provided to validate the effectiveness of the sparse identification method and accuracy of the data-driven modeling. For real system applications, those data may not be accessible.

The output variables of the nonlinear dynamic system depend on both the differential equations and the algebraic equations. The output bus voltage corresponding to different DERs mentioned above are shown in Fig. 5. It has been observed that the output voltage values of the true system is very close to the identified voltage values.

It is realistically possible to obtain the value of the state variables X and approximate the value of $\hat{X}$ required for the algorithm. To provide more evidence on the effectiveness of SINDy algorithm in identifying the system's governing equations, the dynamic states of the true model and identified model for the Wind energy is presented in Fig. 6.

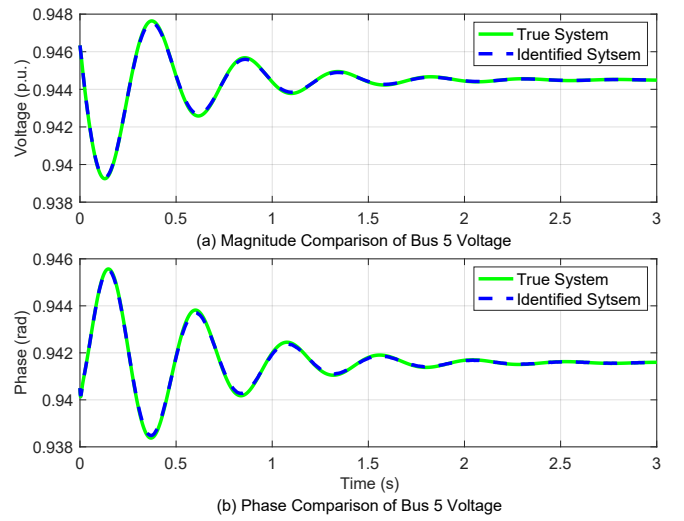


Fig. 5. Comparison of bus voltage magnitude and phase for bus 5.

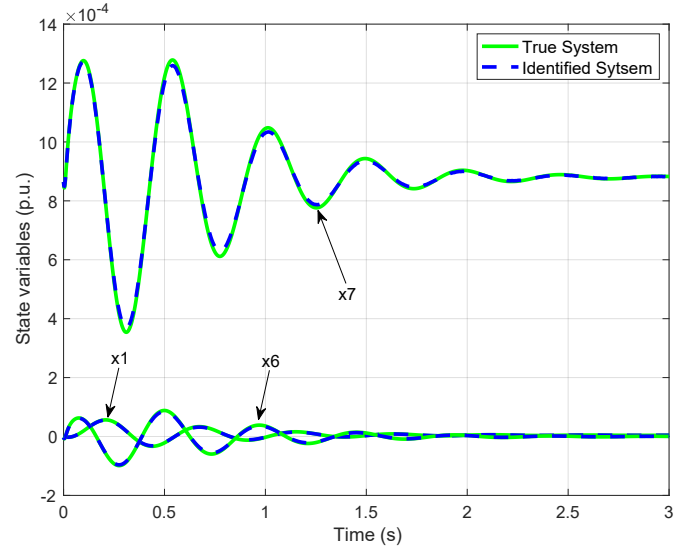


Fig. 6. Comparison of state variables between the true system and identified system for wind energy based DER.

B. Comparison of Eigenvalues

Using the values computed for $\Xi = [\xi_1 \ \xi_2 \ \cdots \ \xi_n]$ as a part of the sparse identification process, the approximate governing equations can be acquired. The obtained equations is a pure ODE form, which have both the state variables and algebraic variables from the true system as unified state variables in the identified system. The approximate governing equations can be linearized around a specific equilibrium point to get the eigenvalues of the linearized model. Fig. 7 shows the eigenvalue comparisons of the true system and the identified system.

From Fig. 7, we can see most of the eigenvalues of the identified system match with the eigenvalues of the true system. The circled dominant eigenvalues primarily determine the

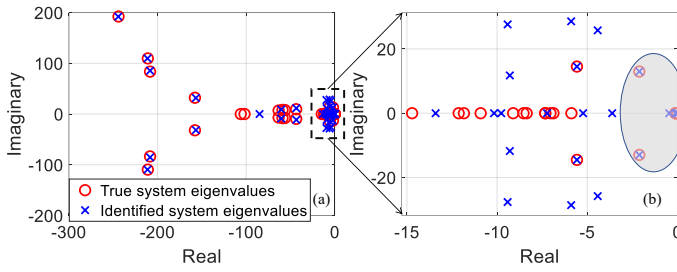


Fig. 7. (a) Comparison of eigenvalues between the true system and the identified system after linearization. (b) Zoom in details of (a)

system's dynamic response; and thus, some small eigenvalue inaccuracies will not cause much difference in the dynamic trajectories. These inaccuracies are attributed to the errors that arise due to the linearization of the nonlinear model, which only describes the dynamics of the nonlinear system around a small operating region.

C. Error Analysis

The actual and identified time-series graph for the state x_1 of the Micro-turbine DER are zoomed in to show the extremely small discrepancy between the true system and the identified system in Fig. 4. To evaluate the identified governing equations, it is imperative to compute the maximum difference between the states of the identified system and the true system. Performing root mean square error calculation on these state variables of the true and identified system, it was found that the error was within the range of $[8.1945e^{-07}, 1.1252e^{-04}]$ with an average value of $4.222e^{-05}$. The maximum error percentage was computed to be $\pm 2.39e^{-03}\%$, which verifies the data-driven model is precise and accurate. To understand the accuracy of the data-driven technique for a large microgrid system, a complex microgrid model with 11 DERs as shown in [1] was simulated and the states were identified using the SINDy method. The error values were found to be in the range of e^{-04} , which verifies that the data-driven model can be applied to identify the governing equations of a large microgrid system without losing the accuracy and precision of the actual system's model.

Constant current load and constant impedance load models were also implemented and the tracking observed showed extreme differences between the system model and the identified model. This inconsistency could be attributed to the fact that the non-linear dynamics in the constant current and impedance load is very less when compared to the constant power load. To identify the accurate dynamic model with constant current and constant impedance load, the algorithm will be modified, making it one of the primary focus for future work on this research topic.

IV. CONCLUSION

Sparse Identification of Non-Linear Dynamics (SINDy) is used in this paper to identify the governing equations of non-linear dynamic microgrid systems. The comparison between the true data measured from the physical system and the data

identified using sparse regression has verified the effectiveness of SINDy in developing data-driven model. It provides a potent tool to analyze the system's transient dynamics when an accurate system's model is unavailable. Since the calculation of SINDy is straightforward, modifications can easily be incorporated into the data-driven model when necessary.

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